

Lecture 12: Midterm Exam 1 (in class)

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- 1 Midterm 1Held in class today, covering Lecture 1 through Lecture 11.: {callout-note icon=false}## Definition: Topics checklist- Probability axioms, conditional probability, Bayes' rule, independence.- Discrete random variables: PMFs, expectation, LOTUS, linearity, variance.- Continuous random variables: PDFs, CDFs, the Normal, joint PDFs.- Derived distributions.- Covariance, correlation, laws of total expectation/variance.- Moment generating functions.:—Good luck! We'll pick up with concentration inequalities on Thursday.## Reading & HomeworkReading: §1-4.5 (cumulative) (Bertsekas & Tsitsiklis, *Introduction to Probability*). 1

—title: “Lecture 12: Midterm Exam 1 (in class)”author: “Jamie Haddock”format: revealjs: output-file: Lecture12_slides slide-number: true preview-links: auto logo: figs/hmc.png css: input/slides.css incremental: true smaller: true code-fold: true embed-resources: true html: code-fold: true embed-resources: true pdf: documentclass: article toc: true number-sections: true geometry: - top=1in - left=1in - bottom=1in - right=1informat-links: falsejupyter: python3filters: - input/remove-pause.luaexecute: echo: true eval: true—

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